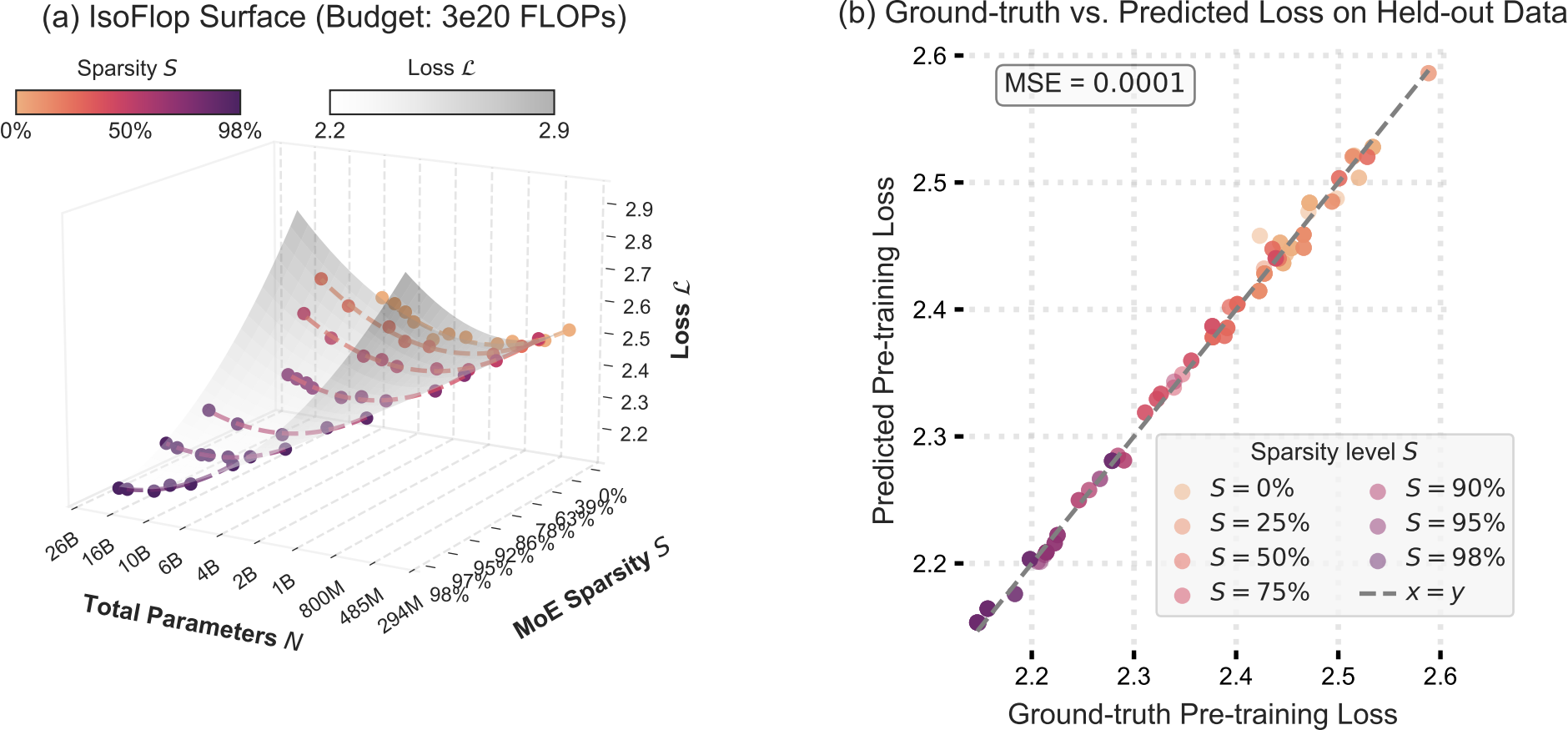
(a) IsoFlop Surface (Budget: 3e20 FLOPs)
Sparsity S
0%
50%
98%
Loss $\mathcal{L}$
2.2
2.9
Loss $\mathcal{L}$
Total Parameters N
26B
16B
10B
6B
4B
2B
1B
800M
485M
294M
MoE Sparsity S
0%
39%
63%
78%
86%
92%
95%
97%
98%
(b) Ground-truth vs. Predicted Loss on Held-out Data
MSE = 0.0001
Predicted Pre-training Loss
Ground-truth Pre-training Loss
Sparsity level S
$S = 0\%$
$S = 25\%$
$S = 50\%$
$S = 75\%$
$S = 90\%$
$S = 95\%$
$S = 98\%$
$x = y$